Matrices & Vectors

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and B be the matrix with rows $\langle -1, 4\rangle$ and $\langle 0, 2\rangle$. Find $A + B$ and $3A$.

$A + B$ has rows $\langle 1, 5\rangle$ and $\langle 3, 1\rangle$. $3A$ has rows $\langle 6, 3\rangle$ and $\langle 9, -3\rangle$.

2 **[Foundational]** • **Target: Addition and scalar multiplication**

Using the same A and B, find $A - 2B$.

$A - 2B$ has rows $\langle 2 - (-2), 1 - 8\rangle = \langle 4, -7\rangle$ and $\langle 3 - 0, -1 - 4\rangle = \langle 3, -5\rangle$.

3 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 2, 1\rangle$, $\langle 3, -1\rangle$) and B (rows $\langle -1, 4\rangle$, $\langle 0, 2\rangle$) from above, find AB.

AB has rows $\langle -2, 10\rangle$ and $\langle -3, 10\rangle$: e.g. row 1, column 1 is $(2)(-1) + (1)(0) = -2$.

4 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and let $\mathbf{v} = \langle 2, -3\rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (2)(2) + (1)(-3), (3)(2) + (-1)(-3)\rangle = \langle 1, 9\rangle$.

5 **[Foundational]** • **Target: Matrix multiplication**

Explain, without computing BA, why AB and BA need not be equal for 2×2 matrices.

Matrix multiplication combines rows of the first factor with columns of the second; swapping the order pairs different rows with different columns, so in general $AB \neq BA$ (matrix multiplication is not commutative).

6 **[Foundational]** • **Target: Determinant and inverse**

Find $\det A$ for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and state whether A is invertible.

$\det A = (2)(-1) - (1)(3) = -5 \neq 0$, so A is invertible.

7 **[Foundational]** • **Target: Determinant and inverse**

Find A^{-1} for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$.

$A^{-1} = \frac{1}{-5} \times$ the matrix with rows $\langle -1, -1\rangle$ and $\langle -3, 2\rangle$, i.e., A^{-1} has rows $\langle 0.2, 0.2\rangle$ and $\langle 0.6, -0.4\rangle$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Determinant and inverse

Without computing it, explain why the matrix with rows $\langle 4, 2\rangle$ and $\langle 6, 3\rangle$ has no inverse.

$\det = (4)(3) - (2)(6) = 12 - 12 = 0$, and a matrix with determinant 0 has no inverse (Theorem 2.1).

9 [Intermediate] • Target: Solving systems with the inverse matrix

Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

M has rows $\langle 3, -1\rangle$ and $\langle 1, 2\rangle$, so $\det M = (3)(2) - (-1)(1) = 7$, and M^{-1} has rows $\langle \frac{2}{7}, \frac{1}{7}\rangle$ and $\langle -\frac{1}{7}, \frac{3}{7}\rangle$. Then $x = \frac{1}{7}[2(8) + 1(5)] = 3$ and $y = \frac{1}{7}[-1(8) + 3(5)] = 1$. Check: $3(3) - 1 = 8$ and $3 + 2(1) = 5$.

10 [Intermediate] • Target: Vector magnitude and dot product

For $\mathbf{u} = \langle 3, 4\rangle$ and $\mathbf{w} = \langle -4, 3\rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

$\mathbf{u} \cdot \mathbf{w} = (3)(-4) + (4)(3) = 0$, and $|\mathbf{u}| = \sqrt{3^2 + 4^2} = 5$.

11 [Intermediate] • Target: Vector magnitude and dot product

What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?

A zero dot product means $\mathbf{u}$ and $\mathbf{w}$ are perpendicular (orthogonal).

12 [Intermediate] • Target: Angle between vectors

Find the angle between $\mathbf{p} = \langle 1, 0\rangle$ and $\mathbf{q} = \langle 1, 1\rangle$.

$\mathbf{p} \cdot \mathbf{q} = 1$, $|\mathbf{p}| = 1$, $|\mathbf{q}| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ$.

13 [Intermediate] • Target: Angle between vectors

Find the angle between $\langle 6, -8\rangle$ and $\langle 3, 4\rangle$, rounded to the nearest tenth of a degree.

Dot product $= 18 - 32 = -14$, magnitudes 10 and 5. $\cos\theta = \frac{-14}{50} = -0.28 \Rightarrow \theta \approx 106.3^\circ$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude and direction angle θ (in degrees, $0^\circ \le \theta < 360^\circ$) of the vector $\mathbf{v} = \langle -5, 5\sqrt{3}\rangle$.

Step 1: Compute magnitude: $|\mathbf{v}| = \sqrt{(-5)^2 + (5\sqrt{3})^2} = \sqrt{25 + 75} = \sqrt{100} = \mathbf{10}$. Step 2: Direction angle: Since $v_x = -5 < 0$ and $v_y = 5\sqrt{3} > 0$, the vector lies in Quadrant II. Reference angle: $\alpha = \tan^{-1}\left(\frac{5\sqrt{3}}{5}\right) = \tan^{-1}(\sqrt{3}) = 60^\circ$. $\theta = 180^\circ - 60^\circ = \mathbf{120}^\circ$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Find the angle θ between the vectors $\mathbf{a} = \langle 1, 2, 2 \rangle$ and $\mathbf{b} = \langle 2, -2, 1 \rangle$.

Step 1: Compute dot product: $\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(-2) + (2)(1) = 2 - 4 + 2 = 0$. Step 2: Conclude angle: Since $\mathbf{a} \cdot \mathbf{b} = 0$ and both vectors have non-zero magnitude ($|\mathbf{a}| = 3, |\mathbf{b}| = 3$): $\cos\theta = 0 \Rightarrow \theta = \mathbf{90}^\circ \quad (\text{or } \frac{\pi}{2} \text{ rad})$.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

Find the inverse of the matrix $A = (4, 7; 1, 2)$.

Step 1: Compute determinant: $\det(A) = (4)(2) - (7)(1) = 8 - 7 = 1$. Step 2: Apply the inverse formula: $A^{-1} = \frac{1}{\det(A)}(d, -b; -c, a) = \frac{1}{1}(2, -7; -1, 4) = (\mathbf{2, -7; -1, 4})$. Check: $AA^{-1} = (8 - 7, -28 + 28; 2 - 2, -7 + 8) = (1, 0; 0, 1) = I$ ✓.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Compute the matrix product AB where $A = (1, 3; -2, 4)$ and $B = (2, 0; 5, -1)$.

Step 1: Compute rows of A dot columns of B: $(AB)_{11} = (1)(2) + (3)(5) = 2 + 15 = 17$ $(AB)_{12} = (1)(0) + (3)(-1) = 0 - 3 = -3$ $(AB)_{21} = (-2)(2) + (4)(5) = -4 + 20 = 16$ $(AB)_{22} = (-2)(0) + (4)(-1) = 0 - 4 = -4$ Conclusion: $AB = (\mathbf{17, -3; 16, -4})$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Calculate the determinant of the matrix $M = (7, 3; -2, 5)$.

Formula: For $(a, b; c, d)$, $\det(M) = ad - bc$. $\det(M) = (7)(5) - (3)(-2) = 35 - (-6) = 35 + 6 = \mathbf{41}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

Find the distance in the polar coordinate plane between the two points $P_1(4, \frac{\pi}{3})$ and $P_2(7, \frac{2\pi}{3})$.

Step 1: Apply the Law of Cosines for polar distance: $d = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2 - \theta_1)}$ Step 2: Substitute values: $r_1 = 4, r_2 = 7, \theta_2 - \theta_1 = \frac{2\pi}{3} - \frac{\pi}{3} = \frac{\pi}{3}$. $d = \sqrt{4^2 + 7^2 - 2(4)(7)\cos\left(\frac{\pi}{3}\right)} = \sqrt{16 + 49 - 56\left(\frac{1}{2}\right)}$ $d = \sqrt{65 - 28} = \sqrt{\mathbf{37}}$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Solve the system using matrix inversion: $4x + 7y = 11$ $x + 2y = 3$

Step 1: Write as matrix equation $A\mathbf{x} = \mathbf{b}$: $(4, 7; 1, 2)(x; y) = (11; 3)$ Step 2: Multiply by $A^{-1} = (2, -7; -1, 4)$ (from Problem 14):

$(x; y) = (2, -7; -1, 4)(11; 3) = (2(11) + (-7)(3); -1(11) + 4(3)) = (22 - 21; -11 + 12) = (\mathbf{1; 1})$.

Thus $\mathbf{x = 1}$ and $\mathbf{y = 1}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Addition and scalar multiplication

Find $\det A$ for the matrix A with rows $\langle -2, 9\rangle$ and $\langle -5, -1\rangle$.

a. 19

b. 47 ✓

c. -43

d. 49

$\det A = (-2)(-1) - (9)(-5) = 2 + 45 = 47$.

22 [Foundational] • Target: Matrix multiplication

Find $\det A$ for the matrix A with rows $\langle 2, -2\rangle$ and $\langle 9, -8\rangle$.

a. 5

b. -34

c. 2 ✓

d. 4

$\det A = (2)(-8) - (-2)(9) = -16 + 18 = 2$.

23 [Foundational] • Target: Determinant and inverse

Find $\det A$ for the matrix A with rows $\langle -4, -8\rangle$ and $\langle -1, -3\rangle$.

a. 20

b. -20

c. 4 ✓

d. 6

$\det A = (-4)(-3) - (-8)(-1) = 12 - 8 = 4$.

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24 [Foundational] • Target: Solving systems with the inverse matrix

Find $\det A$ for the matrix A with rows $\langle 0, -8 \rangle$ and $\langle -4, 4 \rangle$.

a. -30

b. 32

c. -29

d. -32 ✓

$\det A = (0)(4) - (-8)(-4) = 0 - 32 = -32.$

25 [Foundational] • Target: Vector magnitude and dot product

Find $\det A$ for the matrix A with rows $\langle -5, 0 \rangle$ and $\langle -7, 8 \rangle$.

a. -40 ✓

b. 35

c. -37

d. -38

$\det A = (-5)(8) - (0)(-7) = -40 = -40.$

26 [Foundational] • Target: Angle between vectors

Find $\det A$ for the matrix A with rows $\langle 7, -5 \rangle$ and $\langle 6, -4 \rangle$.

a. 2 ✓

b. 4

c. -58

d. 22

$\det A = (7)(-4) - (-5)(6) = -28 + 30 = 2.$

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find $\det A$ for the matrix A with rows $\langle -7, 9 \rangle$ and $\langle 6, -7 \rangle$.

a. -5 ✓

b. 103

c. -3

d. 21

$\det A = (-7)(-7) - (9)(6) = 49 - 54 = -5.$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find $\det A$ for the matrix A with rows $\langle 5, 2 \rangle$ and $\langle -9, -3 \rangle$.

a. -33

b. 3 ✓

c. 5

d. -39

$\det A = (5)(-3) - (2)(-9) = -15 + 18 = 3.$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Let A have rows $\langle -3, 7\rangle$ and $\langle 1, 3\rangle$, and let $\mathbf{v} = \langle 3, -5\rangle$. Find $A\mathbf{v}$.

a. $\langle -42, -12\rangle$

b. $\langle -12, -44\rangle$

c. $\langle -44, -12\rangle$ ✓

d. $\langle -44, -14\rangle$

$A\mathbf{v} = \langle (-3)(3) + (7)(-5), (1)(3) + (3)(-5)\rangle = \langle -44, -12\rangle$.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Let A have rows $\langle 1, 1\rangle$ and $\langle 7, -7\rangle$, and let $\mathbf{v} = \langle 7, 7\rangle$. Find $A\mathbf{v}$.

a. $\langle 14, 0\rangle$ ✓

b. $\langle 0, 14\rangle$

c. $\langle 14, -2\rangle$

d. $\langle 16, 0\rangle$

$A\mathbf{v} = \langle (1)(7) + (1)(7), (7)(7) + (-7)(7)\rangle = \langle 14, 0\rangle$.

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Let A have rows $\langle -1, 6\rangle$ and $\langle 2, 2\rangle$, and let $\mathbf{v} = \langle -1, 5\rangle$. Find $A\mathbf{v}$.

a. $\langle 33, 8\rangle$

b. $\langle 31, 8\rangle$ ✓

c. $\langle 8, 31\rangle$

d. $\langle 31, 6\rangle$

$A\mathbf{v} = \langle (-1)(-1) + (6)(5), (2)(-1) + (2)(5)\rangle = \langle 31, 8\rangle$.

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Let A have rows $\langle -7, -6\rangle$ and $\langle 8, 4\rangle$, and let $\mathbf{v} = \langle -5, -4\rangle$. Find $A\mathbf{v}$.

a. $\langle 61, -56\rangle$

b. $\langle 59, -56\rangle$ ✓

c. $\langle 59, -58\rangle$

d. $\langle -56, 59\rangle$

$A\mathbf{v} = \langle (-7)(-5) + (-6)(-4), (8)(-5) + (4)(-4)\rangle = \langle 59, -56\rangle$.

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33 [Intermediate] • Target: Addition and scalar multiplication

Let A have rows $\langle -2, -6\rangle$ and $\langle 7, -5\rangle$, and let $\mathbf{v} = \langle -9, -3\rangle$. Find $A\mathbf{v}$.

a. $\langle 38, -48\rangle$

b. $\langle 36, -48\rangle$ ✓

c. $\langle 36, -50\rangle$

d. $\langle -48, 36\rangle$

$A\mathbf{v} = \langle (-2)(-9) + (-6)(-3),\ (7)(-9) + (-5)(-3)\rangle = \langle 36, -48\rangle$.

34 [Intermediate] • Target: Matrix multiplication

Let A have rows $\langle -7, -2\rangle$ and $\langle 1, 0\rangle$, and let $\mathbf{v} = \langle 8, -1\rangle$. Find $A\mathbf{v}$.

a. $\langle -54, 6\rangle$

b. $\langle 8, -54\rangle$

c. $\langle -52, 8\rangle$

d. $\langle -54, 8\rangle$ ✓

$A\mathbf{v} = \langle (-7)(8) + (-2)(-1),\ (1)(8) + (0)(-1)\rangle = \langle -54, 8\rangle$.

35 [Intermediate] • Target: Determinant and inverse

Let A have rows $\langle 3, -2\rangle$ and $\langle -5, 1\rangle$, and let $\mathbf{v} = \langle 5, 5\rangle$. Find $A\mathbf{v}$.

a. $\langle 5, -22\rangle$

b. $\langle 7, -20\rangle$

c. $\langle 5, -20\rangle$ ✓

d. $\langle -20, 5\rangle$

$A\mathbf{v} = \langle (3)(5) + (-2)(5),\ (-5)(5) + (1)(5)\rangle = \langle 5, -20\rangle$.

36 [Intermediate] • Target: Solving systems with the inverse matrix

Find A^{-1} for the matrix A with rows $\langle 3, 1\rangle$ and $\langle -1, 0\rangle$.

a. $\langle 0, -1\rangle, \langle 1, 3\rangle$ ✓

b. $\langle 0, 1\rangle, \langle -1, 3\rangle$

c. $\langle 1, 3\rangle, \langle 0, -1\rangle$

d. $\langle 3, 1\rangle, \langle -1, 0\rangle$

$\det A = (3)(0) - (1)(-1) = 1$. $A^{-1} = \frac{1}{1} \times$ the matrix with rows $\langle 0, -1\rangle$ and $\langle 1, 3\rangle$, giving rows $\langle 0, -1\rangle$ and $\langle 1, 3\rangle$.

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37 [Intermediate] • Target: Vector magnitude and dot product

Find A^{-1} for the matrix A with rows $\langle 1, -2\rangle$ and $\langle 2, -5\rangle$.

a. $\langle 5, 2\rangle, \langle -2, -1\rangle$

b. $\langle 2, -1\rangle, \langle 5, -2\rangle$

c. $\langle 5, -2\rangle, \langle 2, -1\rangle$ ✓

d. $\langle 1, -2\rangle, \langle 2, -5\rangle$

$\det A = (1)(-5) - (-2)(2) = -1$. $A^{-1} = \frac{1}{-1} \times$ the matrix with rows $\langle -5, 2\rangle$ and $\langle -2, 1\rangle$, giving rows $\langle 5, -2\rangle$ and $\langle 2, -1\rangle$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve the system $6y = -30$ and $6x = 24$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle -5, 4\rangle$

b. $\langle 6, -5\rangle$

c. $\langle 4, -5\rangle$ ✓

d. $\langle 4, -7\rangle$

M has rows $\langle 0, 6\rangle$ and $\langle 6, 0\rangle$, so $\det M = (0)(0) - (6)(6) = -36$. Solving gives $x = 4$ and $y = -5$. Check: $(0)(4) - 30 = -30$ and $(6)(4) = 24$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Limits

Solve the system $x + 3y = -15$ and $x + 2y = -8$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 6, -7\rangle$ ✓

b. $\langle -7, 6\rangle$

c. $\langle 8, -7\rangle$

d. $\langle 6, -9\rangle$

M has rows $\langle 1, 3\rangle$ and $\langle 1, 2\rangle$, so $\det M = (1)(2) - (3)(1) = -1$. Solving gives $x = 6$ and $y = -7$. Check: $(1)(6) - 21 = -15$ and $(1)(6) - 14 = -8$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Complex Numbers

Solve the system $x - y = 0$ and $-3x - 6y = -54$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 9, 6 \rangle$

b. $\langle 6, 4 \rangle$

c. $\langle 6, 6 \rangle$ ✓

d. $\langle 8, 6 \rangle$

M has rows $\langle 1, -1 \rangle$ and $\langle -3, -6 \rangle$, so $\det M = (1)(-6) - (-1)(-3) = -9$. Solving gives $x = 6$ and $y = 6$. Check: $(1)(6) - 6 = 0$ and $(-3)(6) - 36 = -54$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

For $\mathbf{u} = \langle -20, -21 \rangle$ and $\mathbf{v} = \langle 8, -15 \rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{155}{493}$ ✓

b. $-\frac{155}{493}$

c. $\frac{156}{493}$

d. $\frac{155}{46}$

$\mathbf{u} \cdot \mathbf{v} = (-20)(8) + (-21)(-15) = 155$. $|\mathbf{u}| = 29$, $|\mathbf{v}| = 17$. $\cos\theta = 155/493 = \frac{155}{493}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Advanced Functions & Equations

For $\mathbf{u} = \langle 8, 15 \rangle$ and $\mathbf{v} = \langle 7, 24 \rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{417}{425}$

b. $\frac{208}{21}$

c. $-\frac{416}{425}$

d. $\frac{416}{425}$ ✓

$\mathbf{u} \cdot \mathbf{v} = (8)(7) + (15)(24) = 416$. $|\mathbf{u}| = 17$, $|\mathbf{v}| = 25$. $\cos\theta = 416/425 = \frac{416}{425}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

For $\mathbf{u} = \langle -20, -21 \rangle$ and $\mathbf{v} = \langle 8, -15 \rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{155}{493}$ ✓

b. $-\frac{155}{493}$

c. $\frac{156}{493}$

d. $\frac{155}{46}$

$\mathbf{u} \cdot \mathbf{v} = (-20)(8) + (-21)(-15) = 155$. $|\mathbf{u}| = 29$, $|\mathbf{v}| = 17$. $\cos\theta = 155/493 = \frac{155}{493}$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Limits

If A is a 3×2 matrix and B is a 2×4 matrix, what are the dimensions of the product matrix AB?

a. 3×4 ✓

b. 2×2

c. 4×3

d. Multiplication is undefined

Inner dimensions match ($2 = 2$). The resulting matrix has dimensions equal to outer indices: 3×4.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Complex Numbers

For what value of x is the matrix $[[x, 3], [12, x]]$ singular?

a. ± 6 ✓

b. ± 4

c. ± 12

d. 0

Singular means determinant equals zero: $x^2 - 36 = 0 \Rightarrow x^2 = 36 \Rightarrow x = \pm 6$.

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